Further developments of generalized Jeffreys's approximate Bayes factor

One-sided eJAB, regularity and Equally Weighted Decision Error Profiles

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Abstract

We further develop and exposit generalized Jeffreys's approximate Bayes factor (eJAB; Velidi and Wei, 2025). Equally Weighted Decision Error Profiles compare eJAB with test-statistic Bayes factors (TSBF; Johnson, 2005; Johnson et al., 2023), BIC, and NHST significance rules. For 2×2 and 3×3 Pearson chi-squared independence tests, eJAB generally outperforms estimated-ω TSBF outside small-sample or weak-signal regimes; the trend reverses for 5×5 tables. With true ω, eJAB generally matches or underperforms TSBF. BIC differences are small for single-parameter tests; eJAB gains at weaker effects in multiparameter tests, whereas BIC benefits at stronger effects. Against $p < .05$, eJAB gains with larger samples and stronger signals; weak signals and some small-sample settings favor NHST. We derive a null-centered one-sided eJAB via the Marsman–Wagenmakers correction and weaken regularity conditions while retaining consistency. Three clinical examples illustrate significant findings favoring the null.

1 The unrestricted eJAB definition

For testing $\mathcal{H}_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0$ against $\mathcal{H}_1 : \boldsymbol{\theta} \neq \boldsymbol{\theta}_0$, define

$$eJAB_{01} = \sqrt{n} \exp\left\{ -\frac{1}{2} \left(1 - n^{-1/q}\right) Q_{\chi^2_q}(1-p) \right\}, \qquad eJAB_{10} = eJAB_{01}^{-1}. \tag{1}$$

Here q is the dimension of the test, n is the sample-size quantity appropriate to that test, p is its p-value, and $Q_{\chi^2_q}$ is the chi-squared quantile function. Values greater than one favor the null. The factor $1 - n^{-1/q}$ is retained in this unrestricted definition.

For unrestricted eJAB, the generalized unit-information prior is

$$\boldsymbol{\theta} \mid \mathcal{H}_1 \sim \mathcal{N}_q\Big(\widehat{\boldsymbol{\theta}}, n^{1/q} \mathbf{J}_n^{-1}(\widehat{\boldsymbol{\theta}})\Big), \tag{2}$$

where $\widehat{\boldsymbol{\theta}}$ is the maximum likelihood estimate and $\mathbf{J}_n$ is the observed information in the full sample. With an asymptotic Gaussian posterior and compatible nuisance-parameter priors, the Savage–Dickey density ratio gives the approximate Bayes-factor interpretation (Mulder et al., 2022). Setting the covariance multiplier to $n^{1/q}$ gives the $\sqrt{n}$ prefactor independently of q.

2 One-sided construction under a null-centered prior

2.1 Prior specification and Gaussian approximation

For a scalar directional test, consider

$$\mathcal{H}_0 : \theta = \theta_0 \qquad \text{versus} \qquad \mathcal{H}_+ : \theta > \theta_0. \tag{3}$$

Start with an encompassing unrestricted alternative $\mathcal{H}_1$, and let $v_n = I_n^{-1}(\widehat{\theta})$ denote the inverse observed information in the full sample. The construction used here takes the prior to be centered at the null:

$$\theta \mid \mathcal{H}_1 \sim \mathcal{N}(\theta_0, nv_n), \qquad \theta \mid \boldsymbol{y}, \mathcal{H}_1 \mathrel{\dot\sim} \mathcal{N}(\widehat{\theta}, v_n). \tag{4}$$

The second expression is the leading Gaussian posterior approximation. The prior location is θ_0; the approximate posterior location is $\widehat{\theta}$.

Under the Savage–Dickey representation, with the required compatibility of nuisance-parameter priors, the scalar Gaussian density ratio becomes

$$\widetilde{BF}_{01}^{(0)} = \frac{\phi(\theta_0; \widehat{\theta}, v_n)}{\phi(\theta_0; \theta_0, nv_n)} = \sqrt{n}\exp\left\{-\frac{1}{2}W\right\}, \tag{5}$$
$$W = \frac{(\widehat{\theta} - \theta_0)^2}{v_n}.$$

Here $\phi(\,\cdot\,; \mu, v)$ denotes a normal density with mean μ and variance v. The tilde identifies the Gaussian approximation, and the superscript identifies the null-centered construction. This is the unrestricted base used to obtain the directional approximation below.

2.2 Restriction to the predicted direction

The symmetric encompassing prior assigns probability $1/2$ to $\theta > \theta_0$. For a moderate sample size and a relatively vague prior, the one-sided p-value p_1 gives the approximate posterior mass in that direction (Marsman and Wagenmakers, 2017; Wagenmakers, 2022):

$$\Pr(\theta > \theta_0 \mid \mathcal{H}_1) = \frac{1}{2}, \qquad \Pr(\theta > \theta_0 \mid \boldsymbol{y}, \mathcal{H}_1) \approx 1 - p_1. \tag{6}$$

The encompassing-prior identity and Bayes-factor composition give

$$BF_{+1} = \frac{\Pr(\theta > \theta_0 \mid \boldsymbol{y}, \mathcal{H}_1)}{\Pr(\theta > \theta_0 \mid \mathcal{H}_1)} \approx 2(1 - p_1), \qquad BF_{0+} = \frac{BF_{01}}{BF_{+1}}. \tag{7}$$

Reconstruct the two-sided value as $p = 2\min\{p_1, 1-p_1\}$ and substitute $Q_{\chi^2_1}(1-p)$ for W. The resulting one-sided eJAB is

$$\boxed{eJAB_{0+} = \frac{\sqrt{n}\exp\left\{-\frac{1}{2}Q_{\chi^2_1}(1-p)\right\}}{2(1-p_1)}.} \tag{8}$$

The directional factor retains the information about which side of the null is supported by the data. Equation (8) compares the point null in Equation (3) with the directional alternative.

2.3 Finite-sample convention

Equation (8) omits the factor $(n-1)/n$ from the exponent and retains $1/\{2(1-p_1)\}$. This convention follows the null-centered Gaussian/Savage–Dickey calculation in Equation (5); it is an approximation, not an exact finite-sample conjugate Bayes factor.

The unrestricted definition in Equation (1) continues to use $1-n^{-1/q}$, which is $(n-1)/n$ when $q=1$. Its MLE-centered prior in Equation (2) is distinct from the null-centered prior in Equation (4). Thus, the displayed one-sided formula is not obtained by dividing the finite-sample-corrected Equation (1) by $2(1-p_1)$.

3 Regularity conditions and model-selection consistency

The approximate Bayes-factor derivation uses an asymptotically Gaussian posterior and the stated prior. The consistency statement for Equation (1) concerns the behavior of the p-values. For fixed integer $q \geq 1$, the revised conditions are

$$\textbf{R1}': \quad \Pr(p_n \leq \delta \mid \mathcal{H}_0) \longrightarrow C\delta, \quad 0 < \delta \leq U_p, \tag{9}$$

$$\textbf{R2}: \quad p_n \xrightarrow{P} 0, \qquad D_{q,n} = \sqrt{n} \begin{cases} -p_n \log p_n, & q = 1, \\ p_n, & q \geq 2 \end{cases} \xrightarrow{P} 0 \quad \text{under a fixed } \mathcal{H}_1. \tag{10}$$

In R1′, $U_p \in (0,1]$ and $0 < C \leq 1/U_p$ are constants, and the limit holds for every fixed $\delta \in (0, U_p]$. Only left-tail calibration is required; full asymptotic uniformity is the special case $U_p = C = 1$. R2 retains the original rate when $q = 1$, but removes the logarithmic factor for integer $q > 1$. It remains stronger than $p_n \xrightarrow{P} 0$ alone. These updates are recorded in the reviewer-response revision work plan.

Theorem 1 (Consistency of unrestricted eJAB)**.** *For fixed integer $q \geq 1$, under R1′ and the revised R2,*

$$eJAB_{01} \xrightarrow{P} \begin{cases} \infty, & \textit{under } \mathcal{H}_0, \\ 0, & \textit{under the fixed alternative } \mathcal{H}_1. \end{cases} \tag{11}$$

Proof sketch. Write $W_n = Q_{\chi^2_q}(1-p_n)$ and $a_n = 1 - n^{-1/q}$. Under R1′, for any fixed $\delta \in (0, U_p]$,

$$\Pr\Big(W_n > Q_{\chi^2_q}(1-\delta) \mid \mathcal{H}_0\Big) \leq \Pr(p_n \leq \delta \mid \mathcal{H}_0) \longrightarrow C\delta.$$

Taking δ sufficiently small gives $W_n = O_P(1)$. Hence

$$\log eJAB_{01} = \tfrac{1}{2}\log n - \tfrac{1}{2} a_n W_n \xrightarrow{P} \infty. \tag{12}$$

For $q = 1$, the original small-p chi-squared quantile bound is

$$Q_{\chi^2_1}(1-p) > -2\log p - 2\log(-\log p). \tag{13}$$

For $q \geq 2$, stochastic ordering of chi-squared distributions gives the sharper bound

$$Q_{\chi^2_q}(1-p) \geq Q_{\chi^2_2}(1-p) = -2\log p. \tag{14}$$

This follows by writing a χ^2_q variable as a χ^2_2 variable plus an independent nonnegative χ^2_{q-2} variable when $q > 2$; equality holds when $q = 2$. Define $b_q(p) = -p\log p$ for $q = 1$ and $b_q(p) = p$ for $q \geq 2$, so

$D_{q,n} = \sqrt{n}\, b_q(p_n)$. Because $p_n \xrightarrow{P} 0$, the relevant bound applies with probability tending to one. Both cases yield

$$\begin{aligned} 0 < eJAB_{01} &\le \sqrt{n}\,[b_q(p_n)]^{a_n} \\ &= n^{1/(2n^{1/q})} D_{q,n}^{1-n^{-1/q}} \xrightarrow{P} 0. \end{aligned} \tag{15}$$

The last step uses $n^{1/(2n^{1/q})} \to 1$, $D_{q,n} \xrightarrow{P} 0$, and an exponent tending to one. □

3.1 Fixed and local alternatives

A prior centered at the null and a sequence of local alternatives are different specifications. Consider sequences of the form $\theta_n = \theta_0 + c/\sqrt{n}$. For regular tests, these may produce a nondegenerate noncentral chi-squared limit, so p_n need not converge to zero and R2 need not hold. The consistency statement above therefore does not give consistency along all such local sequences.

Johnson's original local-alternative TSBF differs from the later nonlocal-prior Bayes factor function (Johnson, 2005; Johnson et al., 2023). For every fixed nonzero effect-size ordinate, the latter has pointwise model-selection consistency. These distinctions determine which asymptotic claims apply to a given comparison.

4 Equally Weighted Decision Error Profiles

4.1 Definition and decision rules

For each effect size θ and sample size n, compare a simple null with effect zero and a simple alternative with effect θ. Under 0–1 loss, the equally weighted decision error (EWDE) of decision rule d is

$$\mathrm{EWDE}_d(\theta, n) = \frac{\alpha_d(\theta, n) + \beta_d(\theta, n)}{2}, \tag{16}$$

where α_d is the type I error probability under the null and β_d is the type II error probability at the specified alternative. The profiles display the normalized EWDE comparison, defined by

$$\begin{aligned} \Delta E(\theta, n) &= \mathrm{EWDE}_{\mathrm{comparison}}(\theta, n) - \mathrm{EWDE}_{eJAB}(\theta, n), \\ \mathrm{EWDE}_{\mathrm{norm}}(\theta, n) &= \frac{\Delta E(\theta, n)}{M}, \qquad M = \max_{(\theta,n)\in\mathcal{G}} |\Delta E(\theta, n)|. \end{aligned} \tag{17}$$

The grid $\mathcal{G}$ and normalization constant M are specific to each panel. Positive values favor eJAB and negative values favor the comparison rule. The shared color scale is $[-1, 1]$, and the printed M values retain the scale of the underlying EWDE differences. The calculation is pointwise in effect size, rather than integrated over a composite alternative prior.

The profiles use the unrestricted rule $eJAB_{01} \le 1$. Figure 1 compares it with the TSBF rules and $p < .05$. The Bayes-factor threshold of one corresponds to equal prior model probabilities and 0–1 loss. Figure 2 compares eJAB with $p < .005$ and the BIC Bayes-factor rule.

4.2 Test-statistic Bayes-factor comparisons

The chi-squared comparison uses the Bayes factor function of Johnson et al. (2023), given by

$$BFF_{10}(x_n \mid \tau^2) = (1+\tau^2)^{-q/2-1}\left(1+\frac{\tau^2 x_n}{q(1+\tau^2)}\right)\exp\left\{\frac{\tau^2 x_n}{2(1+\tau^2)}\right\}, \tag{18}$$

$$\tau_\omega^2 = \frac{n\omega^2}{q}.$$

Here ω is the standardized effect-size ordinate. The main figure includes the prespecified-effect rule and its estimated-effect sensitivity version for 2×2, 3×3 and 5×5 contingency tables.

Panel A estimates the effect by $\widehat{\omega} = \sqrt{X^2/n}$, so $\widehat{\tau}^2 = X^2/q$. Since $X^2 = Q_{\chi^2_q}(1-p)$, this plug-in estimate is recoverable from the same observed p, n, q inputs as eJAB. Panel B instead evaluates the rule at the true effect for each comparison setting.

For any fixed $\omega > 0$, the displayed Bayes factor is increasing in X^2. The monotone likelihood ratio of the noncentral chi-squared family therefore makes its upper-tail rejection rule uniformly most powerful among X^2-based tests at its attained size. This Neyman–Pearson argument (Fowlie, 2021) concerns power at a specified size. Separately, cutoff one minimizes prior-predictive error under fixed model priors, equal model probabilities and 0–1 loss; it does not guarantee optimal pointwise average error at each true effect size. The plug-in Panel A remains a sensitivity analysis.

For Pearson chi-squared tests of independence in 2×2 and 3×3 tables, eJAB generally has lower EWDE than the estimated-ω TSBF outside small-sample or weak-signal regimes. In 3×3 tables, small samples can favor TSBF even at larger effects. The broad pattern reverses for 5×5 tables: estimated-effect TSBF generally has lower EWDE, with eJAB gains concentrated in a band of weaker effects. When TSBF uses the true ω, eJAB is generally comparable or worse. The differences are not uniformly small near weak signals, and the few regions favoring eJAB do not establish dominance.

4.3 BIC and NHST comparisons

For BIC, the single-parameter comparisons (t-tests, linear regression and logistic regression) show small absolute EWDE differences. In ANOVA, repeated-measures ANOVA and chi-squared independence tests, eJAB's largest gains occur in weaker-effect bands that move toward zero as sample size increases. At stronger effects, where both rules have high power, BIC's lower Type I error can yield lower EWDE; these differences tend to diminish at larger samples.

Against the NHST significance rule $p < .05$, eJAB generally gains with larger samples and sufficiently strong signals. NHST can have lower EWDE in weak-signal bands. Small-sample behavior also depends on dimension: for multiparameter tests, eJAB can gain at intermediate effects while NHST gains at stronger effects. The normalized profiles show relative patterns; each panel's M records the absolute EWDE scale.

4.4 Local-alternative contours

Dashed curves follow $\theta = c/\sqrt{n}$, with $\theta = \omega$ for table comparisons. They hold $\sqrt{n}\,\theta$ fixed and hence preserve alternative test-statistic distributions that depend on n, θ through this quantity. The eJAB threshold also depends on n, so the curves need not have constant normalized EWDE.

5 Clinical-trial illustration

The revised retrospective analysis uses the August 7, 2025 ClinicalTrials.gov snapshot and retains 16,563 primary- and secondary-outcome analyses from 1,790 exact Early Phase I, Phase I and Phase II interventional trials. Combined-phase trials and Phase III and IV trials are excluded. The 427 explicitly one-sided scalar analyses use Equation (8); the remaining 16,136 use the unrestricted definition with its finite-sample correction.

Table 1 transfers the three illustrative cases from the revised manuscript. Each finding is statistically significant while the applicable eJAB favors the null. These examples span Early Phase I, Phase I and Phase II trials.

Trial and phase	Outcome and comparison	n	p	$eJAB$
NCT00494806 Early Phase I	First flatus: rocking vs standard care; t-test	66	$< .05$	1.23
NCT03907683 Phase I	Activity after a weekday move-more message; t-test	80	.037	1.04
NCT00539539 Phase II	CPR fraction: feedback on vs off; linear regression	1,586	.016	2.19

Table 1: Three illustrative clinical-trial contradictions transferred from the revised manuscript. CPR denotes cardiopulmonary resuscitation. For the first case, the eJAB calculation uses the reported bound .05.

This exploratory reanalysis uses reported numerical bounds, such as .05 for $p < .05$, to compute eJAB while retaining inequalities for significance. Bounds can overstate null evidence, so classifications are provisional. Contradictions establish neither a true null nor a Type I error.

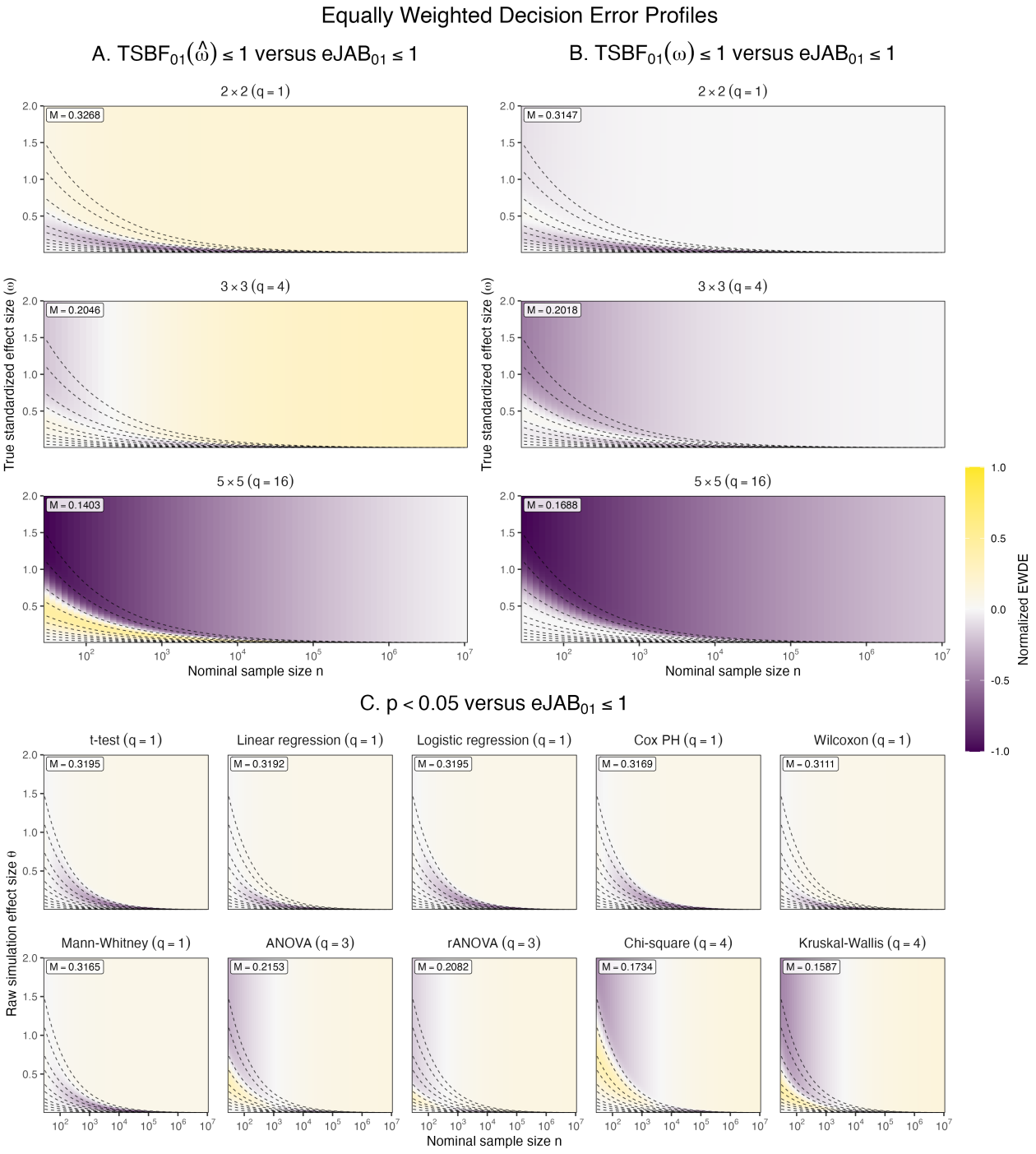


Figure 1: Equally Weighted Decision Error Profiles. (A) $TSBF_{01}(\widehat{\omega}) \leq 1$ and (B) $TSBF_{01}(\omega) \leq 1$, each versus $eJAB_{01} \leq 1$, for 2×2, 3×3 and 5×5 contingency tables. (C) $p < .05$ versus $eJAB_{01} \leq 1$ across ten statistical tests. Colors show normalized EWDE: yellow favors eJAB; purple favors the comparison rule. Each panel reports its normalization constant M. Dashed curves show $\theta = c/\sqrt{n}$, with $\theta = \omega$ in (A) and (B). The figure uses unrestricted eJAB.

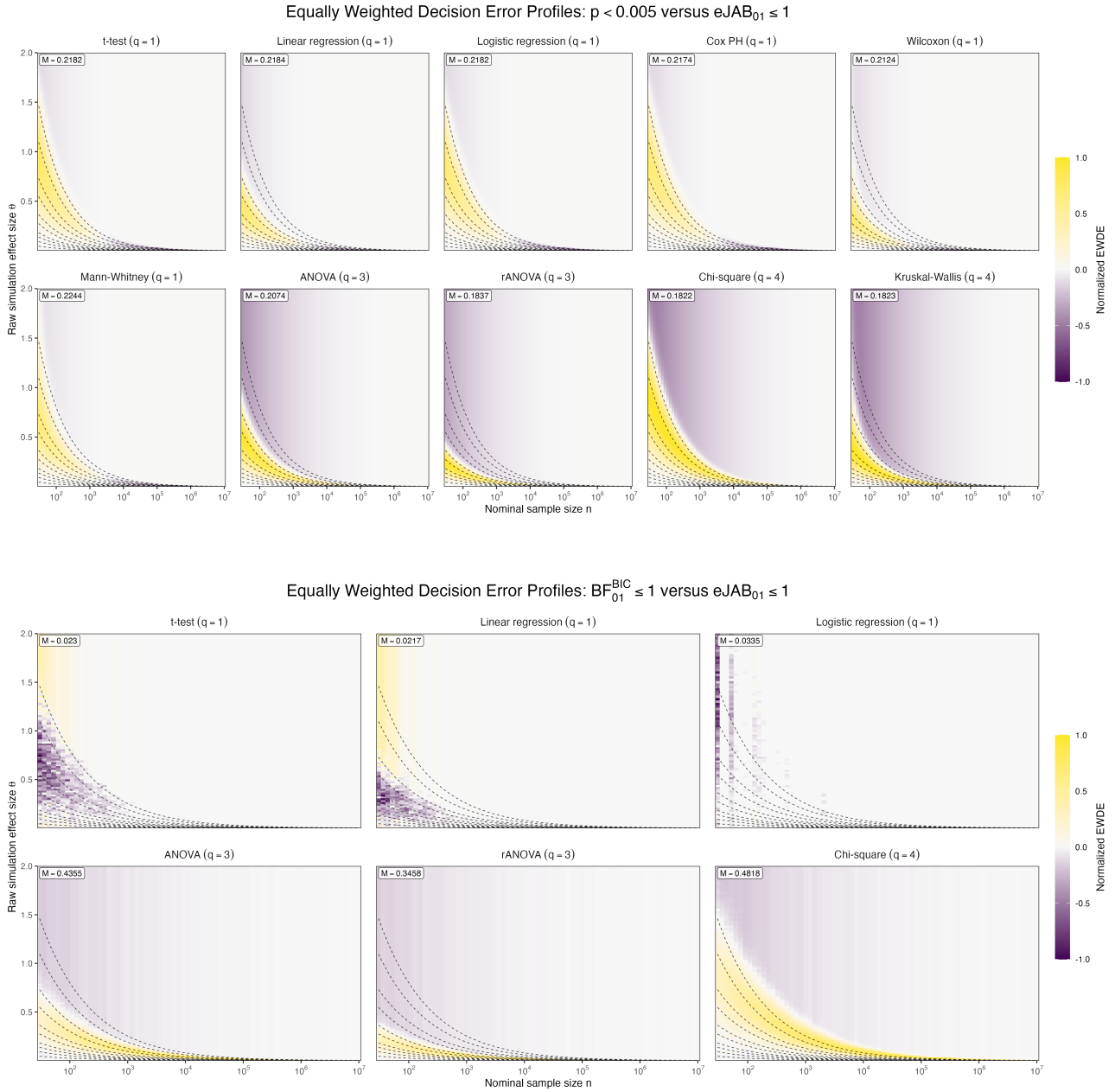


Figure 2: Equally Weighted Decision Error Profiles: additional comparisons. Top: $p < .005$ versus $eJAB_{01} \leq 1$ across ten tests. Bottom: $BF_{01}^{\text{BIC}} \leq 1$ versus $eJAB_{01} \leq 1$ for the six tests with an explicit parametric likelihood. The BIC comparison excludes Cox regression and the three rank tests. Colors show normalized EWDE. Positive values favor eJAB, and negative values favor the comparison rule. The original panel-specific normalization constants are retained.

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